

In [2]: *#Step 1: Import Required Libraries*

```
import pandas as pd
import numpy as np
```

In [5]: *#Step 2: Load the Dataset Properly*

```
# Load dataset
df = pd.read_csv('AirQualityUCI.csv')

# Display first 5 rows
df.head()
```

Out[5]:

	Date	Time	CO(GT)	PT08.S1(CO)	NMHC(GT)	Wind	PT08.S2(NMHC)	Solor.R
--	------	------	--------	-------------	----------	------	---------------	---------

0	10-03-2004	18:00:00	2.6	1360	150	11.9	1046	166
---	------------	----------	-----	------	-----	------	------	-----

1	10-03-2004	19:00:00	2.0	1292	112	9.4	955	103
---	------------	----------	-----	------	-----	-----	-----	-----

2	10-03-2004	20:00:00	2.2	1402	88	9.0	939	131
---	------------	----------	-----	------	----	-----	-----	-----

3	10-03-2004	21:00:00	2.2	1376	80	9.2	948	172
---	------------	----------	-----	------	----	-----	-----	-----

4	10-03-2004	22:00:00	1.6	1272	51	6.5	836	131
---	------------	----------	-----	------	----	-----	-----	-----

In [6]: *#Step 3: Check Dataset Information*

```
# Shape of dataset
print("Shape of Dataset:", df.shape)

# Column names
print("\nColumn Names:")
print(df.columns)

# Dataset information
print("\nDataset Information:")
df.info()
```

Shape of Dataset: (9357, 15)

Column Names:

```
Index(['Date', 'Time', 'CO(GT)', 'PT08.S1(CO)', 'NMHC(GT)', 'Wind',
      'PT08.S2(NMHC)', 'Solor.R', 'PT08.S3(NOx)', 'NO2(GT)', 'PT08.S4(NO
2)',
      'Ozone', 'Temp', 'RH', 'AH'],
      dtype='str')
```

Dataset Information:

```
<class 'pandas.DataFrame'>
```

RangeIndex: 9357 entries, 0 to 9356

Data columns (total 15 columns):

#	Column	Non-Null Count	Dtype
0	Date	9357 non-null	str
1	Time	9357 non-null	str
2	CO(GT)	9357 non-null	float64
3	PT08.S1(CO)	9357 non-null	int64
4	NMHC(GT)	9357 non-null	int64
5	Wind	9357 non-null	float64
6	PT08.S2(NMHC)	9357 non-null	int64
7	Solor.R	9357 non-null	int64
8	PT08.S3(NOx)	9357 non-null	int64
9	NO2(GT)	9357 non-null	int64
10	PT08.S4(NO2)	9357 non-null	int64
11	Ozone	9357 non-null	int64
12	Temp	9357 non-null	float64
13	RH	9357 non-null	float64
14	AH	9357 non-null	float64

dtypes: float64(5), int64(8), str(2)

memory usage: 1.1 MB

```
In [7]: #Step 4: Check Missing Values
# Check missing values
print(df.isnull().sum())
```

```
Date      0
Time      0
CO(GT)    0
PT08.S1(CO)  0
NMHC(GT)  0
Wind      0
PT08.S2(NMHC)  0
Solor.R   0
PT08.S3(NOx)  0
NO2(GT)    0
PT08.S4(NO2)  0
Ozone     0
Temp      0
RH        0
AH        0
dtype: int64
```

```
In [8]: #a DATA CLEANING
#Step 5: Remove Duplicate Rows
# Count duplicate rows
duplicates = df.duplicated().sum()

print("Duplicate Rows:", duplicates)
```

```
# Remove duplicates
df = df.drop_duplicates()

# Check new shape
print("New Shape:", df.shape)
```

Duplicate Rows: 0
New Shape: (9357, 15)

```
In [9]: #Step 6: Handle Missing Values
# Replace -200 with NaN
df.replace(-200, np.nan, inplace=True)
print(df.isnull().sum())
```

```
Date          0
Time          0
CO(GT)        1683
PT08.S1(CO)    366
NMHC(GT)      8443
Wind          366
PT08.S2(NMHC)  366
Solor.R       1639
PT08.S3(NOx)   366
NO2(GT)       1642
PT08.S4(NO2)   366
Ozone         366
Temp          366
RH            366
AH            366
```

dtype: int64

```
In [11]: # Step 7: Convert Date and Time into Datetime Format

# Combine Date and Time columns
if 'Date' in df.columns and 'Time' in df.columns:

    df['Datetime'] = pd.to_datetime(
        df['Date'] + ' ' + df['Time'],
        dayfirst=True
    )

# Display updated dataset
df.head()
```

Out[11]:

	Date	Time	CO(GT)	PT08.S1(CO)	NMHC(GT)	Wind	PT08.S2(NMHC)	Solor.R
0	10-03-2004	18:00:00	2.6	1360.0	150.0	11.9	1046.0	166.0
1	10-03-2004	19:00:00	2.0	1292.0	112.0	9.4	955.0	103.0
2	10-03-2004	20:00:00	2.2	1402.0	88.0	9.0	939.0	131.0
3	10-03-2004	21:00:00	2.2	1376.0	80.0	9.2	948.0	172.0
4	10-03-2004	22:00:00	1.6	1272.0	51.0	6.5	836.0	131.0

In [12]:

```
#Step 8: Statistical Summary
# Statistical summary
df.describe()
```

Out[12]:

	CO(GT)	PT08.S1(CO)	NMHC(GT)	Wind	PT08.S2(NMHC)	Solor.R
count	7674.000000	8991.000000	914.000000	8991.000000	8991.000000	7718.000000
mean	2.152750	1099.833166	218.811816	10.083105	939.153376	246.850000
min	0.100000	647.000000	7.000000	0.100000	383.000000	2.000000
25%	1.100000	937.000000	67.000000	4.400000	734.500000	98.000000
50%	1.800000	1063.000000	150.000000	8.200000	909.000000	180.000000
75%	2.900000	1231.000000	297.000000	14.000000	1116.000000	326.000000
max	11.900000	2040.000000	1189.000000	63.700000	2214.000000	1479.000000
std	1.453252	217.080037	204.459921	7.449820	266.831429	212.970000

In [13]:

```
# b DATA INTEGRATION
# Step 9: Create Environmental Quality Index
```

```

# Create Environmental Index
if {'Temp', 'RH', 'AH'}.issubset(df.columns):

    df['Environmental_Index'] = (
        df['Temp'] + df['RH'] + df['AH']
    ) / 3

# Display data
df.head()

```

Out[13]:

	Date	Time	CO(GT)	PT08.S1(CO)	NMHC(GT)	Wind	PT08.S2(NMHC)	Solor.R
0	10-03-2004	18:00:00	2.6	1360.0	150.0	11.9	1046.0	166.0
1	10-03-2004	19:00:00	2.0	1292.0	112.0	9.4	955.0	103.0
2	10-03-2004	20:00:00	2.2	1402.0	88.0	9.0	939.0	131.0
3	10-03-2004	21:00:00	2.2	1376.0	80.0	9.2	948.0	172.0
4	10-03-2004	22:00:00	1.6	1272.0	51.0	6.5	836.0	131.0

In [14]:

```

#Step 10: Create Integrated Dataset
# Select important columns
selected_columns = []

for col in ['Datetime', 'CO(GT)', 'NMHC(GT)', 'NO2(GT)',
            'Temp', 'RH', 'AH', 'Environmental_Index']:

    if col in df.columns:
        selected_columns.append(col)

# Create integrated dataset
integrated_df = df[selected_columns]

# Display integrated dataset
print(integrated_df.head())

```

	Datetime	CO(GT)	NMHC(GT)	NO2(GT)	Temp	RH	AH	\
0	2004-03-10 18:00:00	2.6	150.0	113.0	13.6	48.9	0.7578	
1	2004-03-10 19:00:00	2.0	112.0	92.0	13.3	47.7	0.7255	
2	2004-03-10 20:00:00	2.2	88.0	114.0	11.9	54.0	0.7502	
3	2004-03-10 21:00:00	2.2	80.0	122.0	11.0	60.0	0.7867	
4	2004-03-10 22:00:00	1.6	51.0	116.0	11.2	59.6	0.7888	

	Environmental_Index
0	21.085933
1	20.575167
2	22.216733
3	23.928900
4	23.862933

```
In [15]: #Step 11: Save Cleaned and Integrated Dataset
# Save final dataset
integrated_df.to_csv('Cleaned_Integrated_AirQuality.csv', index=False)

print("Dataset saved successfully!")
```

Dataset saved successfully!

In []: